Math 514, F19 **Name:** Answer
Quiz 7

Instructions: You must show all of your work to receive credits for correct answers.
Problem 1. (3 points) Find the rate of return for an investment that for an initial payment of 100 return 50 at the end of 1 year and additional 60 at the end of 2 years.

$$P(r) = \frac{50}{1+r} + \frac{60}{(1+r)^2} - 100 = 0$$

let $\beta = \frac{1}{1+r} \Rightarrow 60\beta^2 + 50\beta - 100 = 0$

$$\Rightarrow \beta = \frac{-50+\sqrt{50^2-4\times 60\times(-100)}}{120} \approx \frac{-50+162.788}{120} \approx 0.9399$$

Thus $r = \frac{1}{\beta} - 1 \approx 0.06394$

Problem 2. (4 points) Suppoose that you put \$1,000 in a bank account at time $t = 0$. The varying interest rate is $r(t) = 0.02(2 + \sin t)$ (t measured by year) and compounded continuously. Compute the average of the spot interest rate up to time t, $\tilde{r}(t)$, and the amount in your account after 1 year.

$$\tilde{r}(t) = \frac{1}{t}\int_0^t r(s)\,ds = \frac{1}{t}\int_0^t 0.02(2+\sin s)\,ds$$

$$= \frac{0.02}{t}\left[2s - \cos s\right]_{s=0}^{s=t}$$

$$= \frac{0.02}{t}(2t - \cos t + 1)$$

$$D(t) = D(0)\,e^{\tilde{r}(t)\cdot t} = 1000\,e^{0.02(2t-\cos t+1)}$$

$$\Rightarrow D(1) = 1000\cdot e^{0.02(2\times 1-\cos(1)+1)} \approx 1000\cdot e^{0.04919}$$

$$\approx 1050.42$$

Problem 3. (3 points) Suppose you pay 10 to buy an option to purchase a share of stock at a fixed price $K = 100$ after 2 years. Assume the nominal annual interest rate is 6% and compounded continuously. Find the present value of your return from this investment if the price of the stock after 2 year is 120.

$$PV = (120-100)e^{-rt} - 10$$

$$= 20\cdot e^{-0.06\times 2} - 10$$

$$\approx 17.738 - 10 = 7.738$$